

BUSINESS CONTEXT

Turtle Games is a company that both manufactures and sells its own product, while also sourcing and retailing products made by other companies. Its products include books, board games, video games, and toys. The objective of the data analyst team is to improve overall sales performance through data analysis and customer reviews collected by Turtle Games. The analysis provided attempts to answer some questions posed by Turtle Games:

1. What influences 'loyalty points'?
2. How can customers be clustered for marketing purposes?
3. How can social data be leveraged once customer reviews are collected?
4. Can prescriptive statistics demonstrate the suitability of the 'loyalty points' data to create predictive models?

ANALYTICAL APPROACH

The dataset consisted of 2000 observations and 11 variables. I conducted my analysis using Python and RStudio. I began with Python by importing the data frame provided by Turtle Games and focused my attention on 'loyalty points', 'age', 'remuneration', and 'spending score' features. The data was clean; no null values were found.

First steps:

- NumPy and Pandas were employed to work with arrays
- statistical models were used to obtain estimates and run performance tests
- Matplotlib was used for visualization
- Sklearn was used for statistical modelling

I chose Sklearn. Model_Selection model to create three regression which can put 'loyalty points' feature in relation to the other three variables.

The second step was to import mean_squared_error from Sklearn to predict training and testing sets using the decision tree regressor. The first model overfitted the train set so there was a risk that it might not generalize to new data in the future. I decided to apply some pruning to the model, and the result was still a good fit, avoiding the problem of overfitting.

The third step was to import Seaborn and start the clustering process. I used Re and WordCloud for the sentiment analysis. I made various attempts looking for the best fit. Regarding the sentimental analysis the first step was to delete

duplicate values. I plotted the most frequently used words and then tokenized the 'reviews' and 'summary' features and plotted their polarity

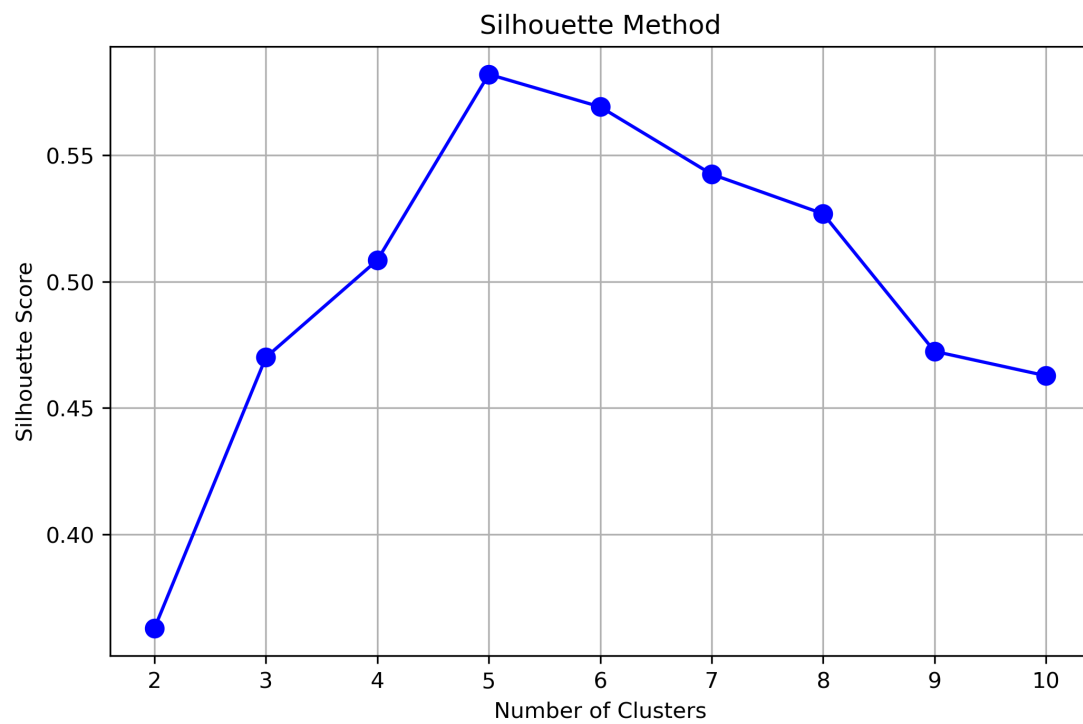
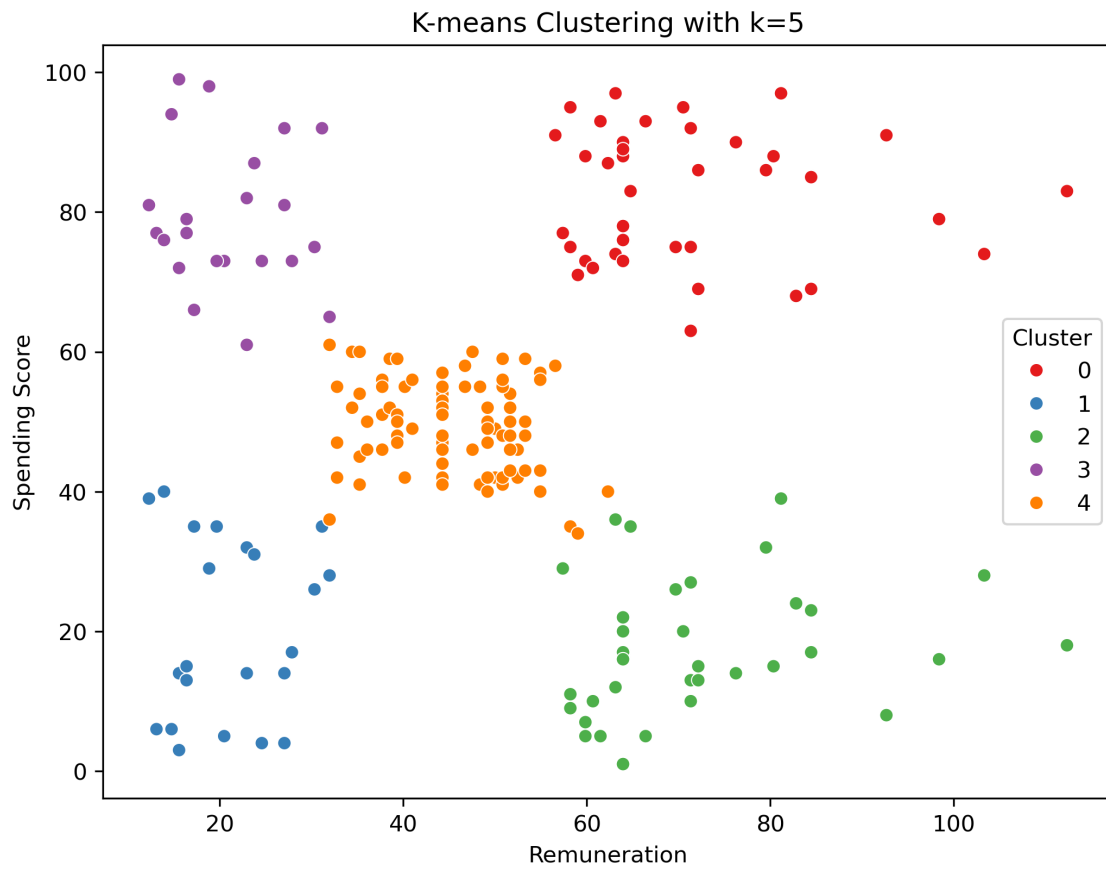
In the end, I chose to analyze the top 20 positive and the top 20 negative reviews and summaries.

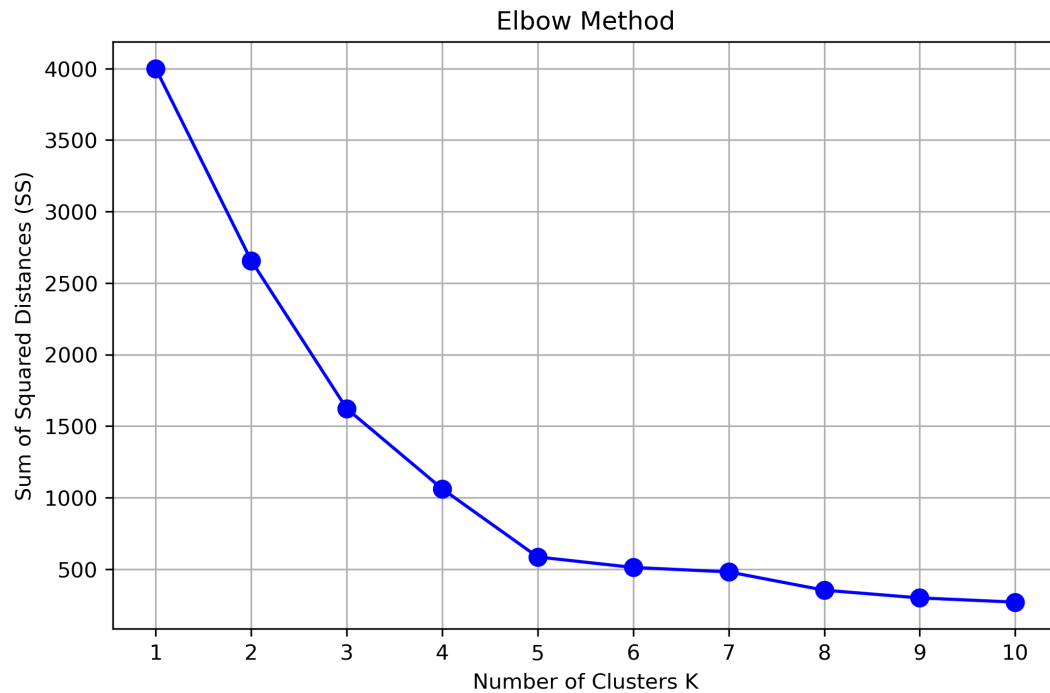
In the analysis I conducted with RStudio I used library as Skimr to summarize the datasets and ggplot for data visualization. After overviewing the dataset, I decided to work with 'age', 'remuneration', 'spending score', 'gender', 'education' and of course 'loyalty points'. I decided to create a visual distribution in histograms, boxplot and scatterplot. Subsequently, I used descriptive statistics to confirm the visual analysis and gain more knowledge regarding data distribution. Initially, I built a multiple linear regression using only spending score and remuneration because the statistical report and the single linear regression suggested that 'age' was not that valuable. In the multi linear regression however, there was a slight improvement, so I decided to integrate the 'age' variable. Finally, I checked the residual for normal distribution plotting to validate the normality assumption of the model.

VISUALIZATION AND INSIGHTS

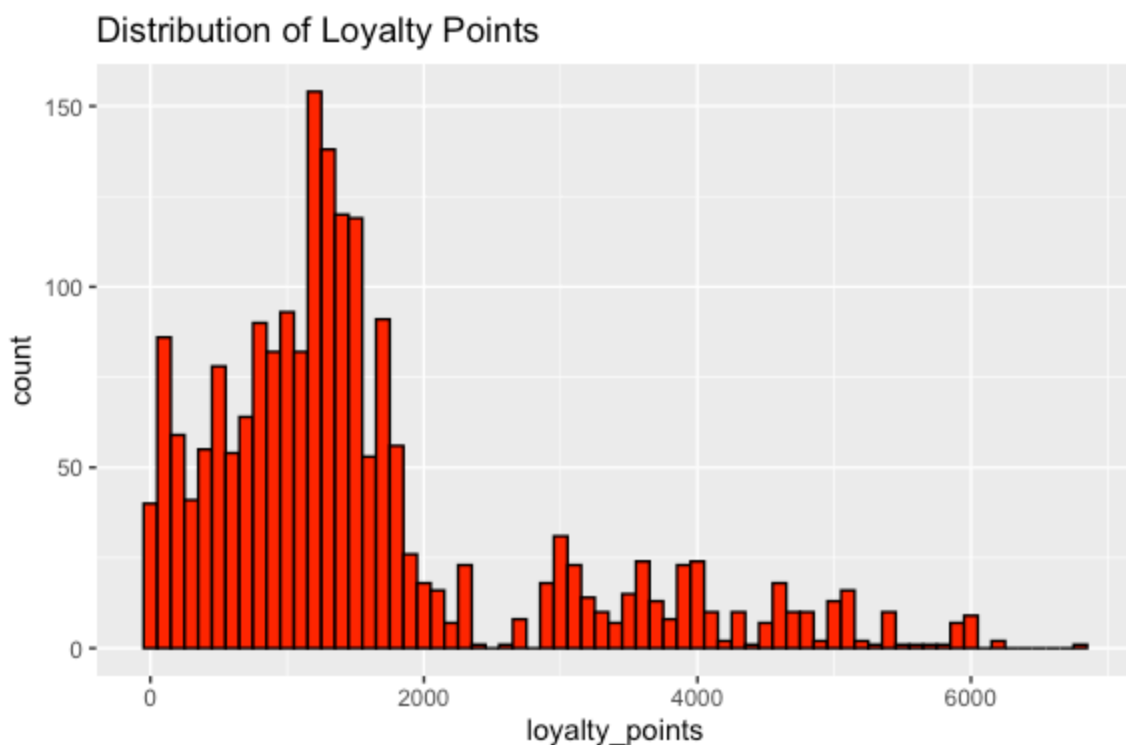
To reach the goals of the analysis, I chose visualizations that highlight variable distribution and offer useful information for marketing purposes.

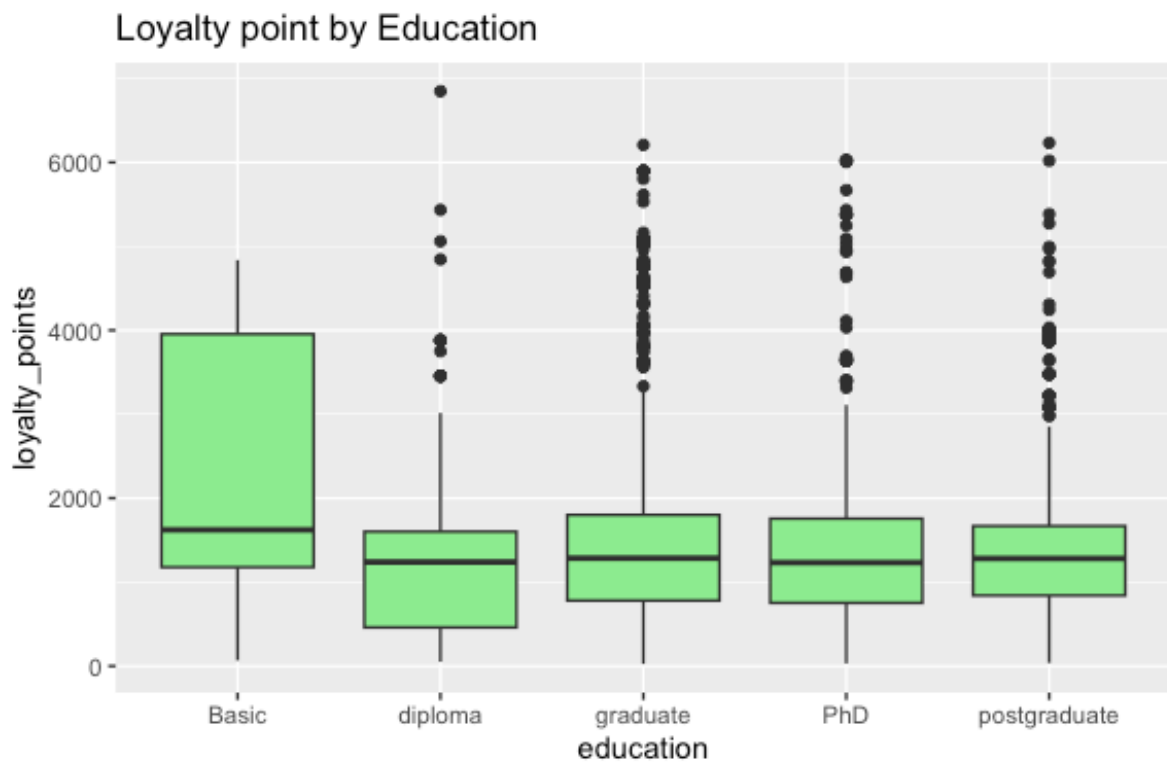
For clustering purposes, I used silhouette methods, elbow methods, histograms, and scatterplot methods. The first two methods helped in selecting the 5 clusters in a clear customer segmentation. This analysis was crucial in clustering customers from high-income high-spending to low-income low-spending providing the marketing department instruments for targeted personalized incentives. It is clear from the three graphics that 5 is the best clustering number.



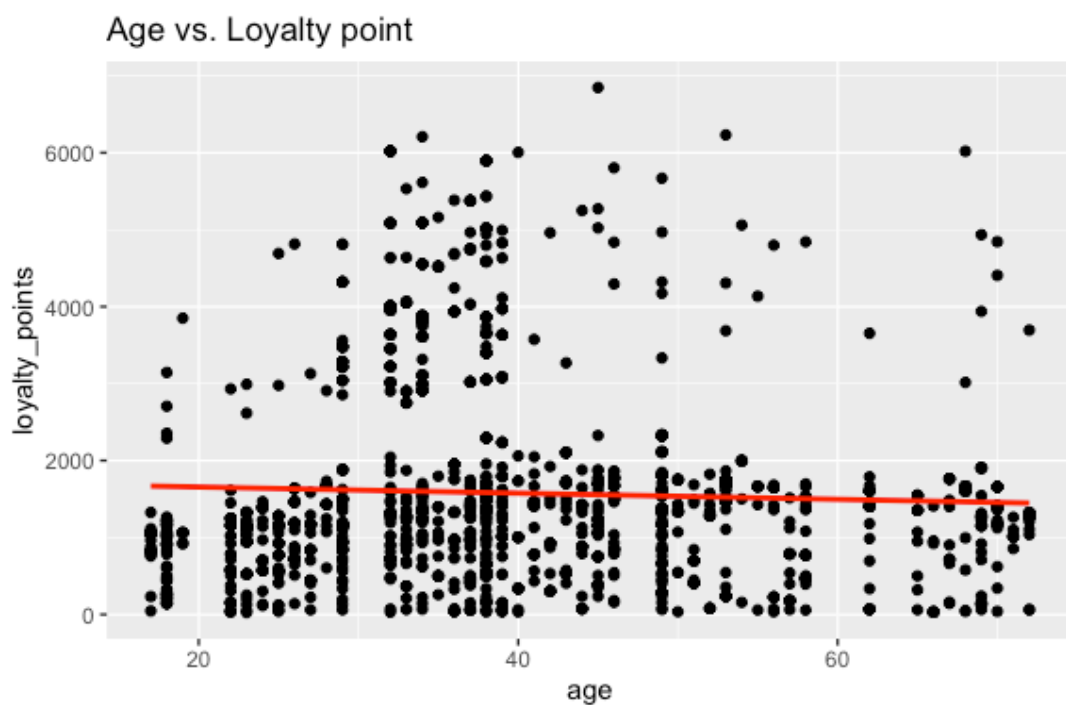


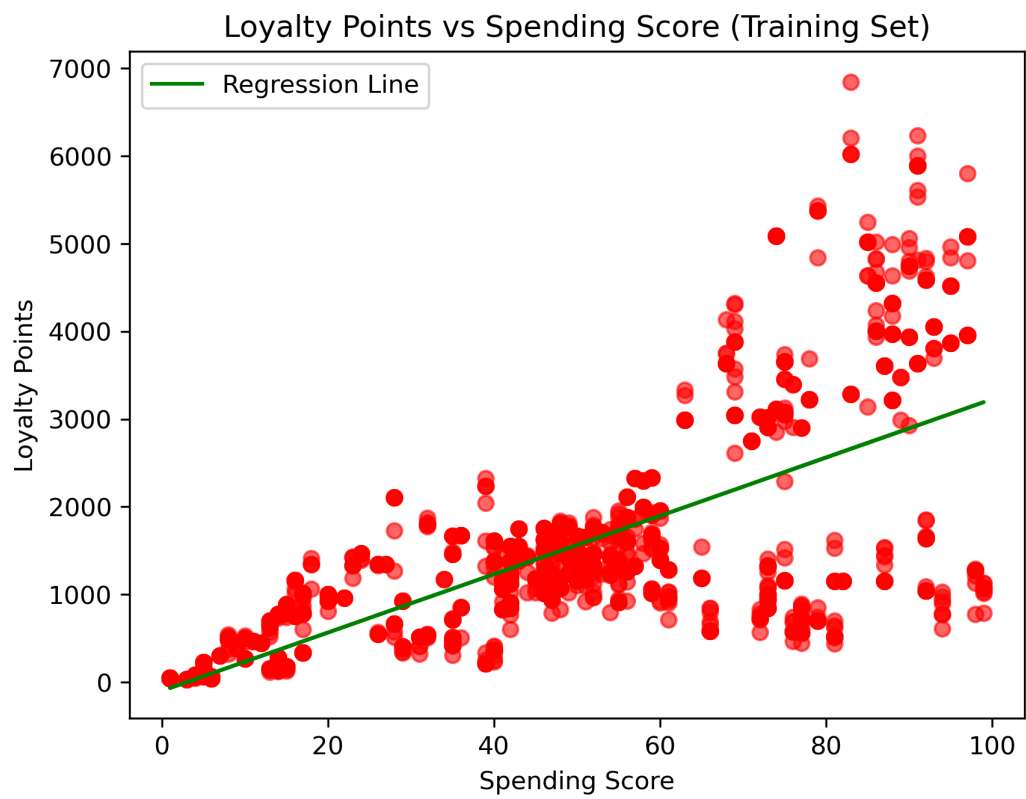
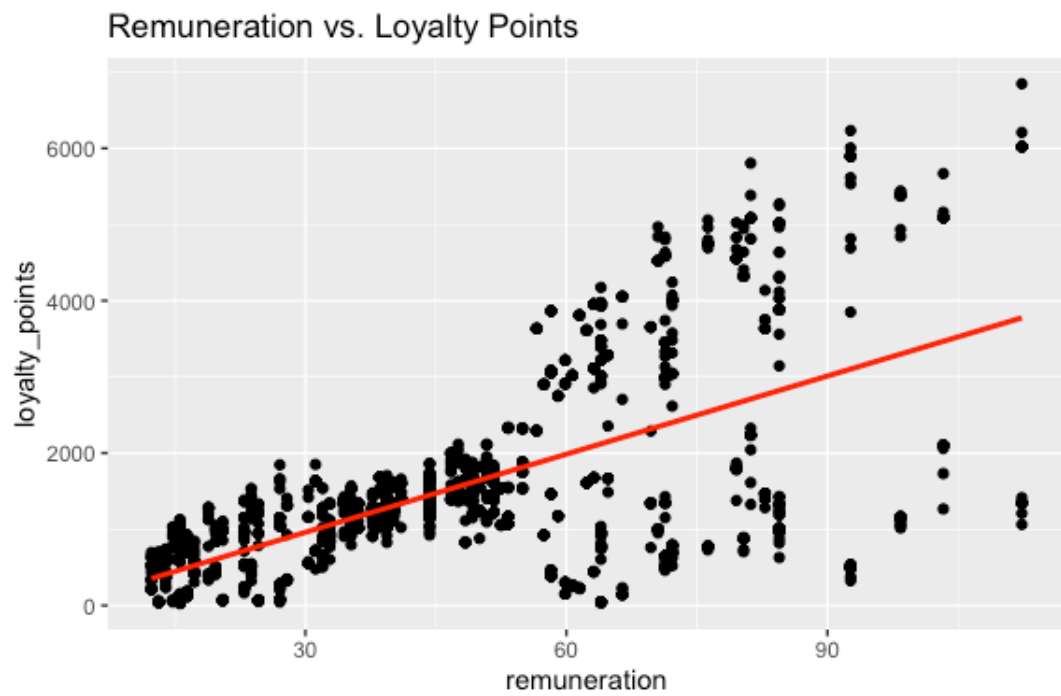
I chose histogram and box plots to investigate the distributions of the variables. The 'loyalty points' histogram showed a right-skewed distribution, where most customers accumulate relatively low points, while a few outliers earn a vast amount. This pattern is also evident in the box plot where the outliers are mostly well-educated.



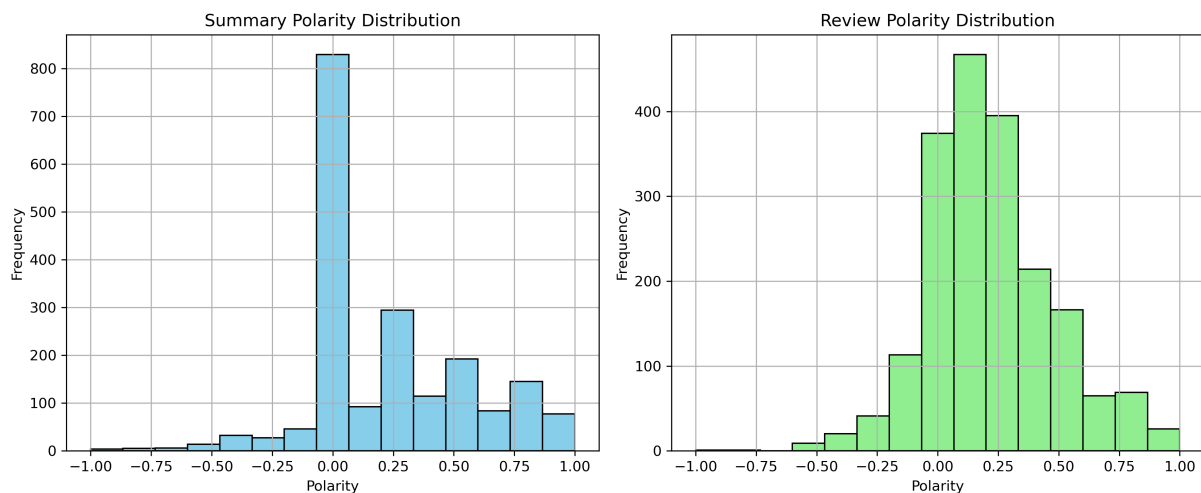


To improve the analysis, I used scatterplot and a regression line to visualize the relationship between these three numerical variables. I detected a positive correlation between 'loyalty points' and income (0.62), and between 'loyalty points' and spending score (0.67). While there was a visually undetectable correlation between age and 'loyalty points' (-0.04). Surprisingly, there was a slightly negative correlation between age and spending score.



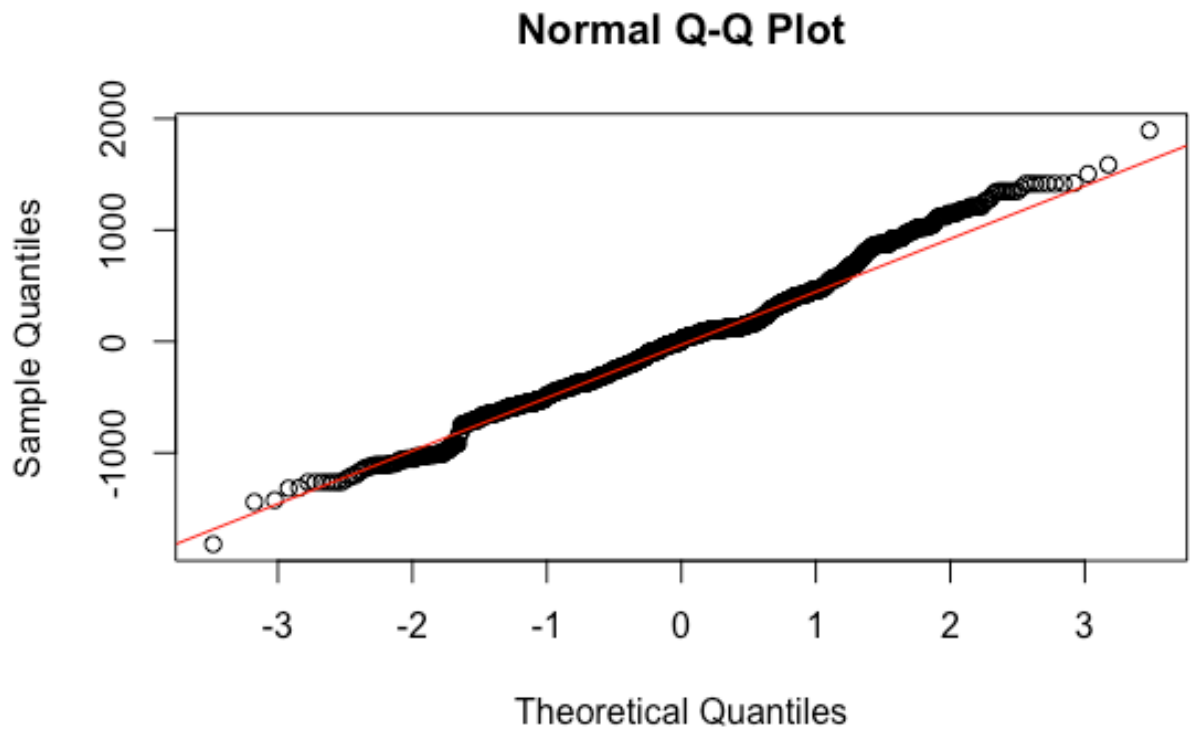
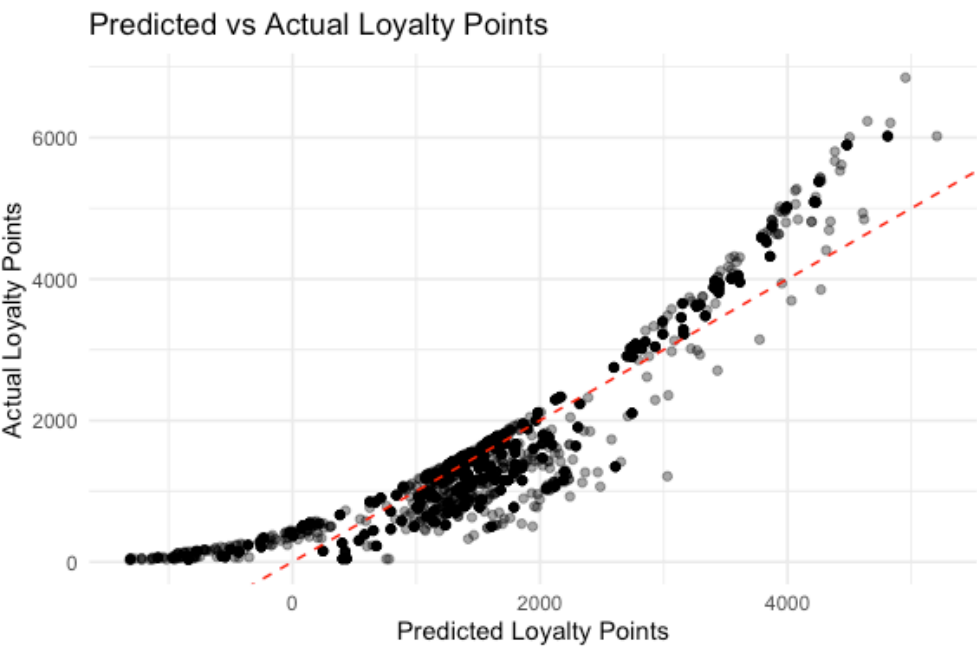


Regarding the qualitative insights in the sentiment analysis, I used words clouds and polarity histograms. The word clouds provided a quick visual overview of the most frequent terms, such as ‘stars’, ‘five’, and ‘great’, which suggest generally positive feedback. The polarity histograms further revealed that both the ‘summary’ and ‘reviews’ variables exhibited predominantly positive sentiment. As the histograms shows, the left-skewed distribution indicates that positive sentiments outweigh negative sentiment.



The multi linear regression model built can predict 83.99% of the variance. This model can support marketing decision by enabling the segmentation of customers into five clusters, helping to identify underperforming groups and design targeted incentive strategies. In the second graph below, I examined the normality of the residuals to validate the model's assumptions. As shown, the residuals are generally distributed around the red line, suggestion that the

normality assumption and that a log-linear transformation may not be necessary to achieve a better fit.



PATTERNS AND PREDICTIONS

The analysis revealed some interesting insights. ‘Loyalty points’ are strongly and positively influenced by both spending score and remuneration, indicating that wealthier customers tend spend more and, consequently, accumulate higher ‘spending scores’ and ‘loyalty points’ features. Although age shows only a weak correlation with ‘loyalty points’, its inclusion in the multiple linear regression model slightly improves the model’s performance. The decision tree model also demonstrates a strong fit, correctly classifying new data approximately 91% of the time.

The multi linear model explains 83.99% of the variance. This model can support marketing decision by segmenting customers into five groups and designing incentive strategies for each segment:

- high both remuneration and spending
- low both remuneration and spending
- high remuneration, low spending
- low remuneration, high spending
- medium both remuneration and spending

The sentiment analysis shows a higher frequency of positive sentiments compared to negative ones, suggesting that most customer comments express satisfaction rather than disappointment. This indicates that overall brand reputation remains strong. However, the presence of frequent negative terms such as “boring” and “disappointing” in both the summary and review texts highlights specific areas where improvements could be made.